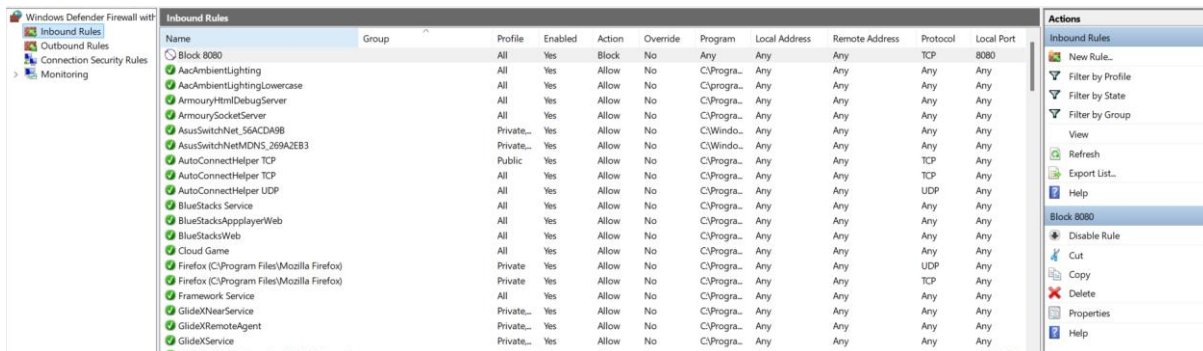


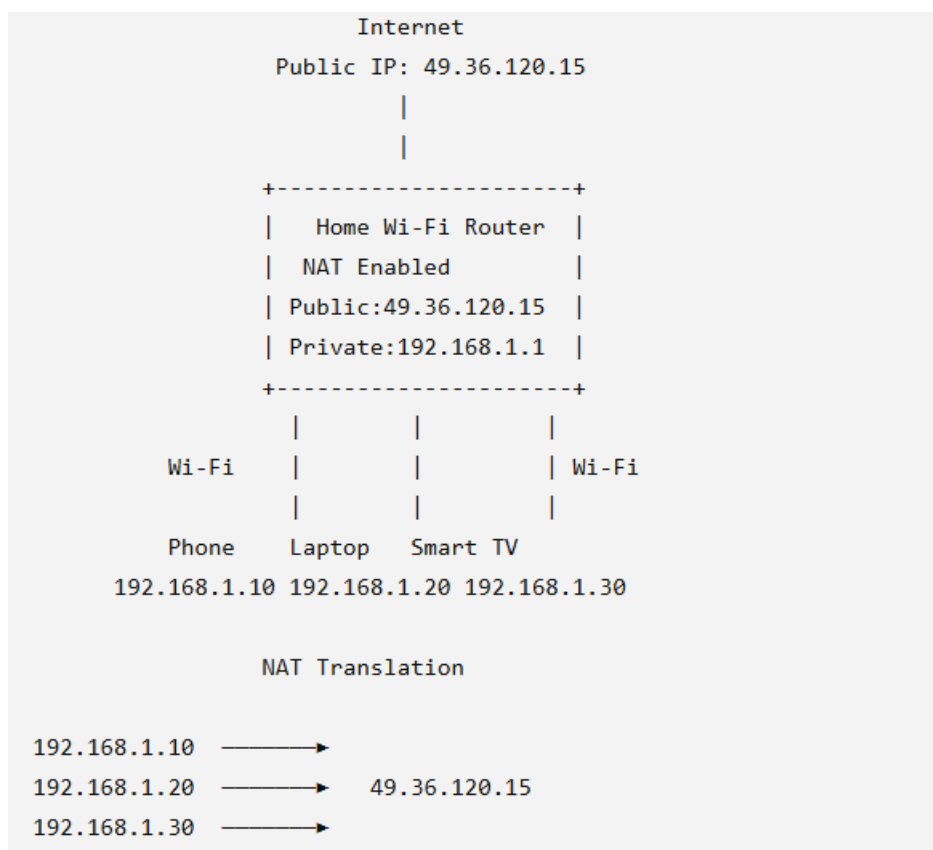
Secure networks with firewalls NAT , Port Filtering , Packet filtering and Other methods

Task 1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



Task 2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.



Task 3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Floating

DMZ

HENETV6

LAN

NS1VPN

TEST

WAN

WG

WLAN

W_GUEST

W_PSK

W_ROKU

IPsec

OpenVPN

WireGuard

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0 / 324.30 MiB	*	*	*	LAN Address	8443 22	*	*		Anti-Lockout Rule	
☐	0 / 0 B	IPv4+6 TCP	LAN net	*	*	25 (SMTP)	*	none		Block 25 outbound	
☒	15 / 210.91 GiB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	
☒	0 / 0 B	IPv6 *	LAN net	*	*	*	*	none			

Add

Add

Delete

Save

Separator

Blocking outbound SMTP (port 25) prevents infected devices from sending unauthorized email directly to external mail servers. This helps reduce spam, limits malware activity, and protects the network from being used in spam campaigns.

Task 4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.

Rule	Action
Allow HTTP (Port 80)	Allow
Allow DNS (Port 53)	Allow
Block SSH (Port 22)	Block

Then the results would be:

Packet	Result	Reason
HTTP request from 192.168.1.5 → 172.217.166.78:80	Allowed	HTTP (Port 80) is permitted
SSH request from 10.0.0.10 → 34.201.23.1:22	Blocked	SSH (Port 22) is blocked
DNS query from 192.168.1.7 → 8.8.8.8:53	Allowed	DNS (Port 53) is permitted

Task 5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

1. Blocking Malicious Traffic

- Firewalls inspect incoming packets and block suspicious requests, such as denial-of-service (DDoS) attacks, unauthorized scans, or traffic from known malicious IP addresses.

2. Restricting Access to Internal Services

- Packet filtering allows only required ports (such as HTTPS on port 443) to be accessible from the internet while blocking administrative ports like SSH or database ports from public access.

3. Web Application Protection

- Web application firewalls (WAFs) filter malicious HTTP/HTTPS requests, helping protect against SQL injection, cross-site scripting (XSS), bot abuse, and other attacks that target web applications.

These firewall and packet-filtering techniques help keep user data secure, maintain service availability, and reduce the risk of unauthorized access or abuse.